LAB 03

PH142 Fall 2025

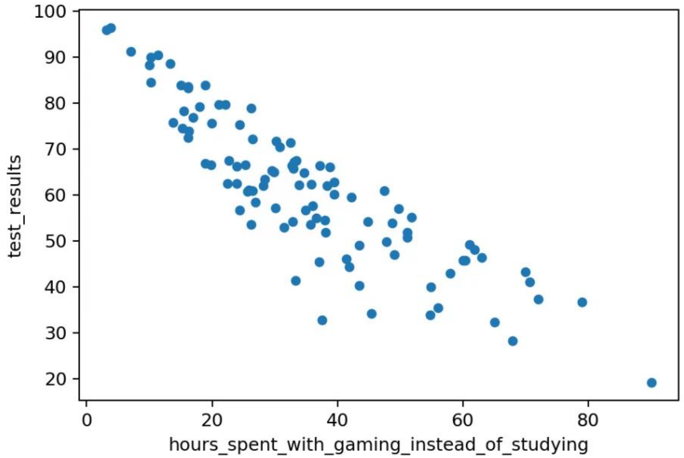
Announcements

- **Lab03:** due Fri 2/6 at 11:59pm
- **Quiz02:** due Sat 2/7 at 12:00pm
    - The quiz is available from **9 AM Friday** to **12 PM Saturday**
    - You have 30 minutes to complete the quiz

Week 3 Lecture Review

Scatterplots

1. Direction - Is it positive or negative?
2. Form - Is it linear or nonlinear?
3. Strength - Is it strong or weak?
4. Outliers - Are there any obvious ones?



How would you interpret this scatterplot?

Can you think of any confounding factors?

Week 3 Lecture Review

Scatterplots

geom_point is used for scatter plots:

Ex.
```
name of plot <- ggplot(data = dataset, aes(x=var, y=var)) +
                geom_point(na.rm = TRUE) +
                theme_minimal(base_size = 15) +
                labs (x= "", y= "", title= "")
```

- To color the points by a variable include col=variable
- To create separate plots for combinations of levels of 2 vars i.e. (gender) use facet_wrap ex. facet_wrap (~ gender)

Week 3 Lecture Review

Correlation

- Correlation measures the **strength** and **direction** of a linear relationship between two quantitative variables
- Also written as **r**
- Takes values between -1 to 1 inclusively

Week 3 Lecture Review

Intro to Linear Regression

- **Regression:** Straight line fitted to data to minimize distance between the data and fitted line
  - "Line of best fit" = **a + bx**
  - **a** = **intercept** (Predictive Value of y when x=0)
  - **b** = **slope** r*(sy/sx)
- Interpretation
  - Intercept : the value of the outcome when x = 0
  - Slope: For a one-unit change in x, the outcome changes by [number] [units]

Week 3 Lecture Review

Intro to Linear Regression

- lm() is the function for a linear model
  - lm(formula = y ~ x, data = your_dataset)
  - Add regression line to a scatterplot using geom_abline()
  - glance(data_lm) will give r squared
- **Interpretation of lm():** A one unit change in **X** is associated with a ____ increase/decrease of **Y**.
- **Interpretation of r-squared**: the fraction of the variation in the values of y that is explained by the line of best fit (the regression of y on x)
- Find **correlation** in data using summarize(corr_variable = cor(var1, var2))

LAB 03 Walkthrough

Lab Submission

- Follow the directions on the LAB03 file
- Submit using the **Terminal Tab** (next to the console in the bottom left pane)
  - Copy and paste the given line into the terminal
  - Follow prompts (NOTE: the terminal will **not** show your password being typed out!)
- <u>**CHECK IN GRADESCOPE THAT ALL YOUR TESTS PASSED**</u>